CDA-EOS: Curvature–Driven Hydrogen-Bond Cooperativity in an Extension of IAPWS-95 for Liquid Water

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Abstract

We introduce the Curvature–Driven Association Equation of State (CDA-EOS) for liquid water. The construction combines three existing ingredients in a single Helmholtz free-energy formulation: (i) the IAPWS-95 Helmholtz equation as a high-accuracy reference for bulk thermodynamic properties, (ii) a SAFT-style association term representing hydrogen bonding via a four-site network, and (iii) thermodynamic curvature in the sense of Ruppeiner, computed directly from the IAPWS-95 free energy, used as a scalar measure of correlation strength. The curvature field $R(T, \rho)$ is mapped to a dimensionless network factor that modulates the effective association strength, yielding a curvature-dependent association contribution on top of the IAPWS reference. All thermodynamic quantities then follow from a single closed-form molar Helmholtz free energy $a_{\mathrm{tot}}(T, \rho)$ by differentiation. Here ρ denotes molar density; IAPWS-95 is evaluated using its standard mass-density variable via $\rho_{\mathrm{mass}} = M_w \rho$. This is a new EOS construction and theoretical framework in the sense that the association term is not fitted independently, but is tied to correlation information already encoded in a standard engineering equation of state. We derive explicit expressions for the CDA-EOS association term, the resulting pressure, the isobaric expansion coefficient, and the qualitative impact on the temperature of maximum density (TMD) line and heat-capacity anomalies. This curvature-feedback mechanism provides a microscopic-style interpretation of water's anomalies within a high-accuracy engineering framework. A preliminary numerical demonstration is included to show base-EOS recovery, direct-metric curvature mapping, stable pressure-level CDA association corrections, and a prototype C_p consistency check on a masked single-phase liquid grid.

1 Introduction

Liquid water exhibits a set of anomalous thermodynamic behaviours that distinguish it sharply from simple liquids. At ambient pressure there is a temperature of maximum density (TMD) near 4°C, and on supercooling both the isothermal compressibility and the isobaric heat capacity rise steeply [5, 4]. These anomalies have often been interpreted in terms of competition between more open, tetrahedral local structures and more compact, high-density

structures, and a possible liquid–liquid critical point in the deeply supercooled regime has been discussed in a wide range of simulation and theoretical studies [4, 3].

For practical thermodynamic calculations, the IAPWS-95 formulation provides a widely used reference equation of state (EOS) for ordinary water. It expresses the Helmholtz free energy in terms of reduced variables and a residual contribution $\phi^{(r)}(\tau, \delta)$ that is fitted to a large body of experimental data over broad ranges of temperature and pressure [1, 2]. From this single Helmholtz function, one can obtain pressure, entropy, internal energy, and response functions for engineering use with high accuracy. IAPWS-95, however, is essentially empirical: it does not expose explicit microscopic degrees of freedom for hydrogen bonding or local structural states.

In contrast, the Statistical Associating Fluid Theory (SAFT) and its variants treat associating fluids by decomposing the Helmholtz free energy into ideal, reference, chain, and association parts, with the association term derived from Wertheim's theory for site–site bonding [6, 7]. For water, SAFT and PC-SAFT models often represent hydrogen bonding through multi-site association schemes and, in some cases, explicit two-state or multi-state descriptions of high-density and low-density local structures [8, 9, 11]. In these models, the association parameters are fitted to reproduce known anomalies such as the TMD and the shape of $C_p(T, p)$.

A third line of work applies thermodynamic geometry to fluid equations of state. In Ruppeiner's formulation, one defines a Riemannian metric from second derivatives of the entropy and computes a scalar curvature R that is related to the correlation volume of the fluid [12, 14]. For several real substances, including water, $R(T, p)$ has been evaluated from experimental or reference EOS and used to identify regimes of large correlations and "solid-like" liquid states [13, 15]. In these studies, the EOS is taken as given and the curvature is used as a diagnostic; it is not typically fed back into the EOS as an active ingredient.

The central aim of this work is to connect these three strands in a minimal, explicit way. We propose the Curvature–Driven Association Equation of State (CDA-EOS) for liquid water, in which:

1. The IAPWS-95 molar Helmholtz free energy $a_{\mathrm{IAPWS}}(T, \rho)$ is used as a fixed base EOS for water in the liquid and supercooled regions.

2. Thermodynamic curvature $R(T, \rho)$ is computed from a_{IAPWS} and mapped to a dimensionless network factor $n(T, \rho)$, which increases with correlation strength.

3. A four-site SAFT-style association term, representing hydrogen bonding, is added on top of a_{IAPWS}, with its effective association strength multiplied by a simple function of $n(T, \rho)$.

This yields a total molar Helmholtz free energy

$$a_{\mathrm{tot}}(T, \rho) = a_{\mathrm{IAPWS}}(T, \rho) + \lambda a_{\mathrm{assoc}}(T, \rho; R), \tag{1}$$

which we refer to as the CDA-EOS. In this sense, the present work introduces a new EOS construction and theoretical framework: the association contribution is not chosen independently of the reference EOS, but is regulated by the correlation structure implied by IAPWS-95 itself. Figure 1 summarizes the construction.

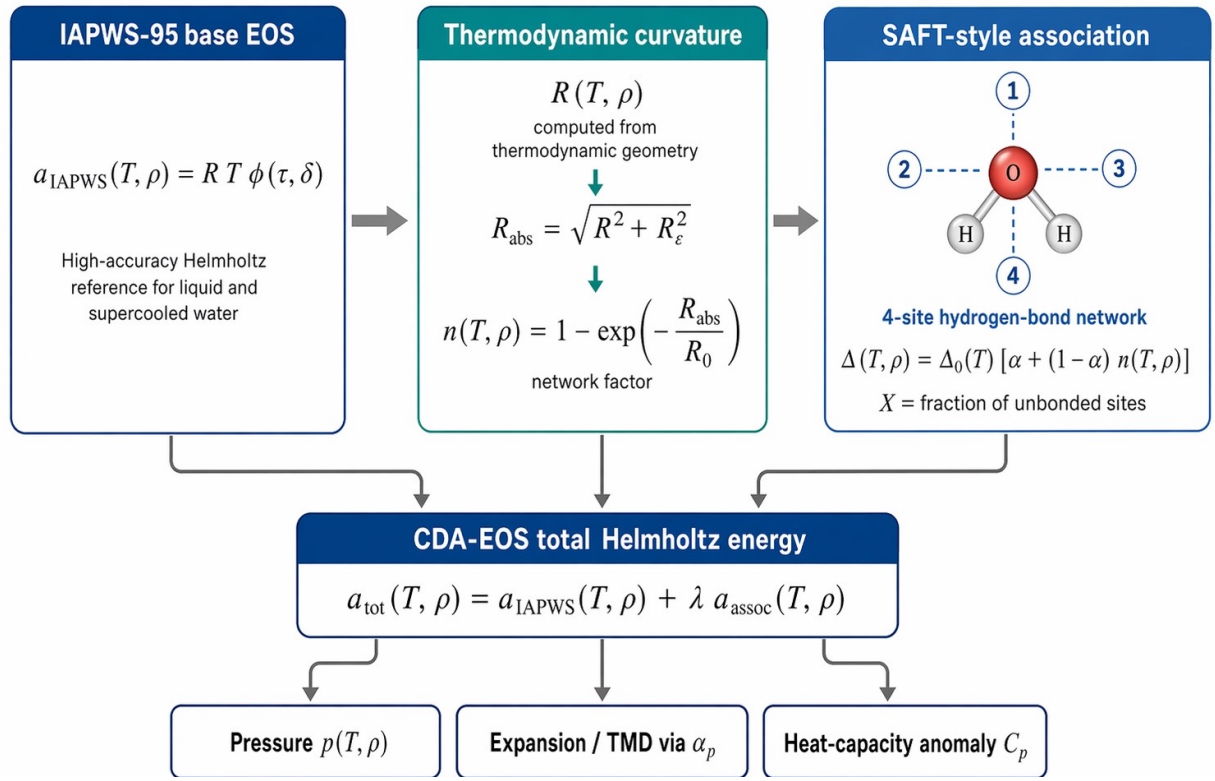

Figure 1: Schematic of the curvature-driven association construction in the proposed CDA-EOS. The IAPWS-95 base Helmholtz energy supplies the macroscopic reference EOS; thermodynamic curvature $R(T,\rho)$ is mapped through the regularised magnitude R_{abs} into a dimensionless network factor $n(T,\rho)$; the resulting factor modulates the SAFT-style association contribution before all thermodynamic properties are derived from $a_{\mathrm{tot}}(T,\rho)$.

The rest of this note sets out the explicit form of the curvature–driven association term, derives expressions for pressure and selected response functions, and discusses how the CDA-EOS could be parameterised and compared with existing two-state and SAFT-type models for water. The intention is not to provide a final, fully fitted EOS, but to define a concrete, closed-form structure that can be implemented on top of existing IAPWS codes and tested against data and simulations in follow-up work.

2 Background

2.1 IAPWS-95 Helmholtz free energy and density convention

Throughout this paper, ρ denotes the molar density (mol/m^3), and $a(T, \rho)$ denotes the molar Helmholtz free energy (J/mol). IAPWS-95 is commonly expressed using the mass density ρ_{mass} (kg/m^3). The two are related by

$$\rho_{\text{mass}} = M_w \rho, \tag{2}$$

where M_w is the molar mass of water.

To avoid ambiguity, we treat IAPWS-95 as providing a Helmholtz free energy in its native variable $a_{\text{IAPWS}}(T, \rho_{\text{mass}})$ and then define the molar-density form used in this paper by composition:

$$a_{\text{IAPWS}}(T, \rho) \equiv a_{\text{IAPWS}}(T, \rho_{\text{mass}})\big|_{\rho_{\text{mass}} = M_w \rho}. \tag{3}$$

Any derivative with respect to ρ therefore uses the chain rule, for example

$$\left(\frac{\partial a_{\text{IAPWS}}}{\partial \rho}\right)_T = M_w \left(\frac{\partial a_{\text{IAPWS}}}{\partial \rho_{\text{mass}}}\right)_T. \tag{4}$$

We use IAPWS-95 in its standard reduced variables

$$\tau = \frac{T_c}{T}, \qquad \delta = \frac{\rho_{\text{mass}}}{\rho_{c,\text{mass}}} = \frac{M_w \rho}{\rho_{c,\text{mass}}}, \tag{5}$$

where T_c and $\rho_{c,\text{mass}}$ are the critical temperature and critical mass density used by IAPWS-95.

The IAPWS-95 formulation expresses the Helmholtz free energy as [1, 2]

$$a_{\text{IAPWS}}(T, \rho_{\text{mass}}) = RT\, \phi(\tau, \delta), \tag{6}$$

with

$$\phi(\tau, \delta) = \phi^{(0)}(\tau, \delta) + \phi^{(r)}(\tau, \delta). \tag{7}$$

Here ϕ is the standard IAPWS-95 dimensionless Helmholtz function, and R is the molar gas constant. This is equivalent to the more common mass-specific form $f = R_{\text{spec}} T \phi$ with $R_{\text{spec}} = R/M_w$ and $a = M_w f$.

Once a is known, all thermodynamic properties follow from standard relations; for example,

$$p(T,\rho) = \rho^2 \left(\frac{\partial a}{\partial \rho}\right)_T, \tag{8}$$

$$s(T,\rho) = -\left(\frac{\partial a}{\partial T}\right)_\rho, \tag{9}$$

$$u(T,\rho) = a + Ts, \tag{10}$$

where s and u are molar entropy and molar internal energy.

2.2 SAFT association free energy (molar form)

In SAFT, the Helmholtz free energy is commonly decomposed as [6, 7]

$$a = a^{\mathrm{id}} + a^{\mathrm{ref}} + a^{\mathrm{chain}} + a^{\mathrm{assoc}}, \tag{11}$$

where a^{assoc} accounts for specific site–site association such as hydrogen bonding. For a single associating species with association sites a, the association term in molar form is

$$\frac{a^{\mathrm{assoc}}}{RT} = \sum_a \left(\ln X_a - \frac{X_a}{2} + \frac{1}{2}\right), \tag{12}$$

where X_a is the fraction of sites of type a that are not bonded. The fractions X_a are determined by mass-action equations of the form

$$X_a = \left[1 + \rho \sum_b X_b \Delta_{ab}(T,\rho)\right]^{-1}, \tag{13}$$

where ρ is molar density (mol/m^3), and Δ_{ab} has units of m^3/mol so that $\rho\Delta_{ab}$ is dimensionless. A typical choice is

$$\Delta_{ab}(T) = K_{ab}\left(e^{\varepsilon_{ab}/(RT)} - 1\right), \tag{14}$$

with parameters K_{ab} (m^3/mol) and ε_{ab} (J/mol). For water, SAFT and PC-SAFT variants with appropriately chosen association schemes have been shown to reproduce many anomalous properties when combined with a two-state picture [8, 9, 11, 10].

2.3 Thermodynamic curvature

Thermodynamic geometry defines a Riemannian metric on the space of equilibrium states. In the entropy representation $s = s(u,v)$, where u is molar internal energy and v is molar volume, the Ruppeiner metric is [12, 14]

$$g_{ij} = -\frac{\partial^2 s}{\partial x^i \partial x^j}, \tag{15}$$

with coordinates $x^1 = u$, $x^2 = v$. From g_{ij} one computes the scalar curvature R, which has been argued to scale with a correlation volume and to change sign across different interaction

regimes [12, 13, 14]. For several simple fluids, including water, $R(T, p)$ has been evaluated along coexistence curves and inside the liquid region, revealing regimes with large negative R (strong attractive correlations) and narrow slabs of positive R interpreted as "solid-like" liquid states [13, 15].

The scalar curvature carries units of volume in the standard fluctuation interpretation. With molar variables, it is natural to treat R as having units of molar volume (m^3/mol), up to a conventional multiplicative factor that depends on the exact coordinate choice. The CDA-EOS construction uses R only through ratios, for example R_{abs}/R_0, so any fixed convention is acceptable as long as it is used consistently.

3 Curvature–Driven Association Model

We now construct a curvature–driven association contribution on top of IAPWS-95.

3.1 Base equation of state

We treat IAPWS-95 as a fixed base molar Helmholtz free energy, evaluated in its native mass-density form and then mapped into the paper's molar-density variable via $\rho_{\text{mass}} = M_w \rho$:

$$a_{\text{IAPWS}}(T, \rho) \equiv a_{\text{IAPWS}}(T, \rho_{\text{mass}})\big|_{\rho_{\text{mass}} = M_w \rho} = RT\phi(\tau, \delta). \tag{16}$$

3.2 Curvature and network factor

From $a_{\text{IAPWS}}(T, \rho)$ we can obtain $u(T, \rho)$ and $s(T, \rho)$ and thus the metric g_{ij} and its scalar curvature $R(T, \rho)$. We do not reproduce those lengthy expressions here; we assume that $R(T, \rho)$ can be computed numerically for any state point using IAPWS-95, following EOS-based curvature evaluations such as Refs. [13, 15].

Because the CDA-EOS uses derivatives of $a_{\text{tot}}(T, \rho)$, it is helpful for the network factor to be smooth even if $R(T, \rho)$ crosses zero. We therefore define a smoothed magnitude

$$R_{\text{abs}}(T, \rho) = \sqrt{R(T, \rho)^2 + R_\epsilon^2}, \tag{17}$$

where $R_\epsilon > 0$ is a small regularisation scale. R_ϵ has the same units as R, and is chosen small enough to only matter very near $R = 0$. We define a dimensionless network factor

$$n(T, \rho) = 1 - \exp\left(-\frac{R_{\text{abs}}(T, \rho)}{R_0}\right), \tag{18}$$

where $R_0 > 0$ sets a scale at which correlations are considered strong. R_0 has the same units as R so that R_{abs}/R_0 is dimensionless. In weakly correlated regions, $R_{\text{abs}} \ll R_0$ and

$$n(T, \rho) \approx \frac{R_{\text{abs}}(T, \rho)}{R_0}, \tag{19}$$

so the boost to association strength is approximately linear in the correlation volume. Near regions of strong correlation, $R_{\text{abs}} \gg R_0$ and $n \to 1$. Because R_{abs} uses the magnitude of

R, the present model deliberately ignores the sign of the curvature and treats the network factor as a proxy for correlation strength only. A sign-sensitive extension could be explored in later work, but is not required for the minimal curvature-feedback construction pursued here. A qualitative example of $n(T)$ along the 1-bar isobar is shown in Fig. 2.

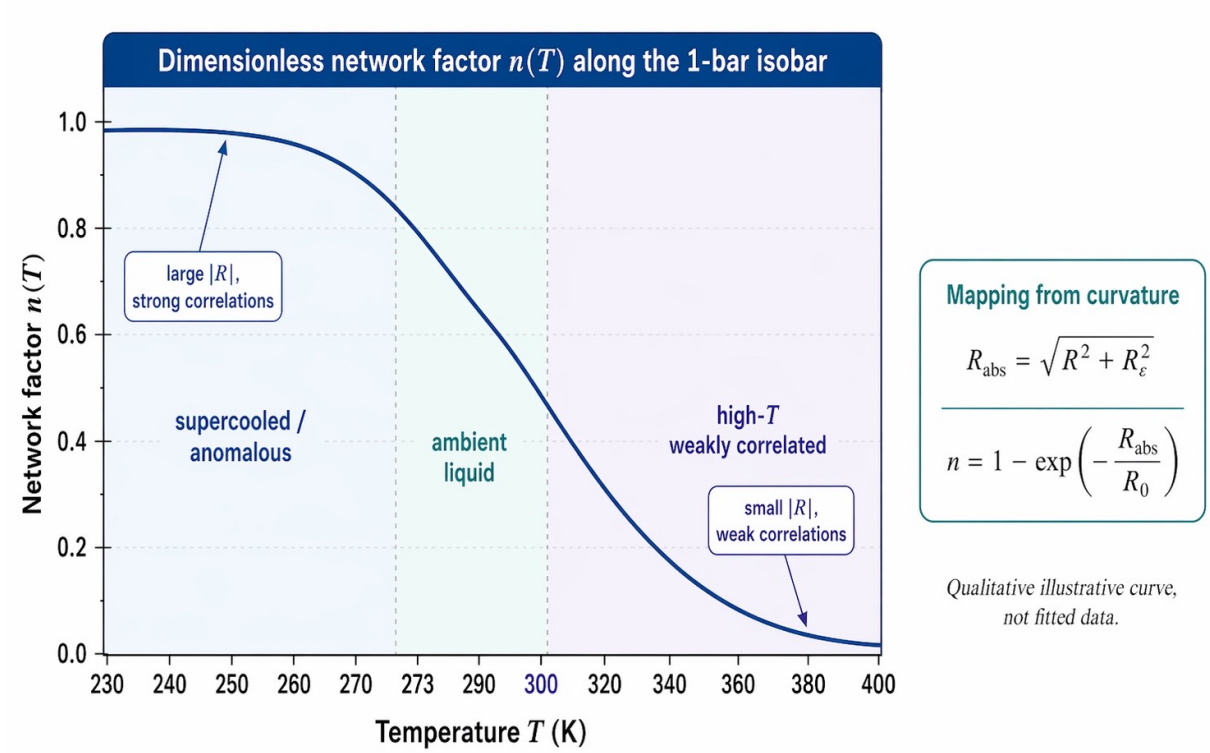

Figure 2: Qualitative behaviour of the dimensionless network factor $n(T)$ along the 1-bar isobar in the proposed CDA-EOS. This is a schematic illustration, not fitted data and not a validated numerical prediction. It shows the intended mapping: large $|R|$ in cold or supercooled, strongly correlated regimes drives n toward unity, while weakly correlated high-temperature states drive n toward zero.

3.3 Curvature-dependent association strength

We model water as a SAFT-like associating fluid with $m = 4$ identical sites (two donors and two acceptors) and a single effective association parameter. The bare association strength (without curvature) is taken as

$$\Delta_0(T) = K_0 \left(e^{\varepsilon/(RT)} - 1\right), \tag{20}$$

where K_0 is a bonding volume (m^3/mol) and $\varepsilon > 0$ is a bonding energy (J/mol). Here K_0 should be read as an effective parameter that can absorb simple reference-structure prefactors, for example a contact g_{ref}, when one wants to compare against standard SAFT conventions. The curvature-dependent association strength is

$$\Delta(T, \rho) = \Delta_0(T) \left[\alpha + (1 - \alpha) n(T, \rho)\right], \tag{21}$$

where $\alpha \in (0,1)$ sets a baseline level of association even in weakly correlated states. When $n \to 0$, $\Delta \to \alpha\Delta_0$; when $n \to 1$, $\Delta \to \Delta_0$.

For a single site type with multiplicity m, the mass-action equation reduces to

$$X = [1 + \rho m X \Delta(T,\rho)]^{-1}, \tag{22}$$

where $X(T,\rho)$ is the fraction of unbonded sites. This yields a quadratic equation

$$(\rho m \Delta) X^2 + X - 1 = 0, \tag{23}$$

with physically relevant solution

$$X(T,\rho) = \frac{-1 + \sqrt{1 + 4\rho m \Delta(T,\rho)}}{2\rho m \Delta(T,\rho)}. \tag{24}$$

The bonded fraction per site is $B(T,\rho) = 1 - X(T,\rho)$. Figure 3 summarizes the four-site association structure and the curvature-dependent bond-strength feedback.

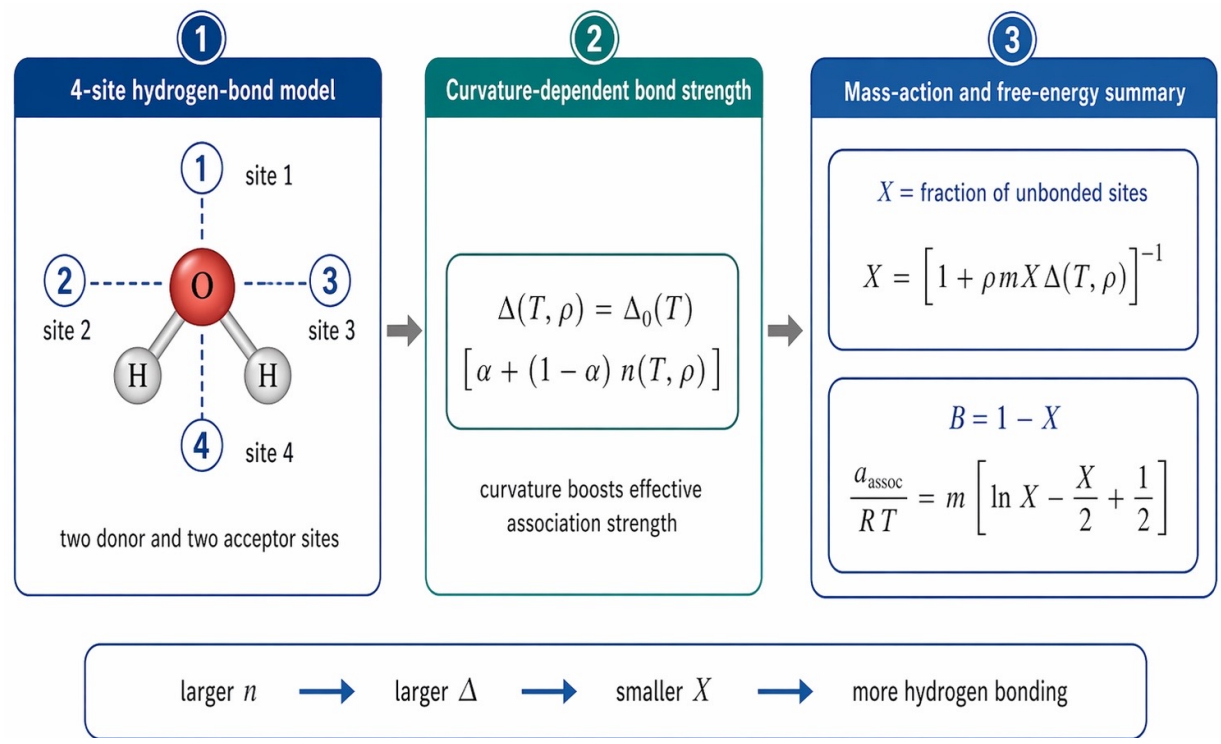

Figure 3: Four-site association model and curvature-dependent bonding variables used in the CDA-EOS. The water molecule is represented by two donor and two acceptor sites, the curvature-driven network factor modifies $\Delta(T,\rho)$, and the resulting mass-action variable X determines the association free-energy contribution.

3.4 Association free energy and total Helmholtz energy

The association contribution in molar form is

$$\frac{a_{\text{assoc}}(T,\rho)}{RT} = m\left(\ln X(T,\rho) - \frac{X(T,\rho)}{2} + \frac{1}{2}\right). \tag{25}$$

We propose the total molar Helmholtz free energy

$$a_{\mathrm{tot}}(T,\rho) = a_{\mathrm{IAPWS}}(T,\rho) + \lambda a_{\mathrm{assoc}}(T,\rho), \tag{26}$$

with λ a dimensionless prefactor intended to reduce double counting of association physics already encoded empirically in the IAPWS residual term. In the simplest implementation one can set $\lambda = 1$ and treat the association term as a structured correction, then adjust $(K_0, \varepsilon, \alpha, R_0, R_\epsilon, \lambda)$ so that macroscopic properties remain consistent with data.

4 Derived Thermodynamic Properties

4.1 Pressure

The pressure is

$$p(T,\rho) = \rho^2 \left(\frac{\partial a_{\mathrm{tot}}}{\partial \rho}\right)_T = p_{\mathrm{IAPWS}}(T,\rho) + p_{\mathrm{assoc}}(T,\rho), \tag{27}$$

with

$$p_{\mathrm{IAPWS}}(T,\rho) = \rho^2 \left(\frac{\partial a_{\mathrm{IAPWS}}}{\partial \rho}\right)_T, \tag{28}$$

and

$$p_{\mathrm{assoc}}(T,\rho) = \lambda \rho^2 \left(\frac{\partial a_{\mathrm{assoc}}}{\partial \rho}\right)_T. \tag{29}$$

Using the explicit form of a_{assoc},

$$\frac{\partial a_{\mathrm{assoc}}}{\partial \rho} = RTm \left(\frac{1}{X} - \frac{1}{2}\right) \left(\frac{\partial X}{\partial \rho}\right)_T, \tag{30}$$

so that

$$p_{\mathrm{assoc}}(T,\rho) = \lambda RTm\rho^2 \left(\frac{1}{X} - \frac{1}{2}\right) \left(\frac{\partial X}{\partial \rho}\right)_T. \tag{31}$$

It can be useful to express $(\partial X/\partial \rho)_T$ in terms of the auxiliary variable

$$A(T,\rho) \equiv \rho m \Delta(T,\rho), \tag{32}$$

so that

$$X(A) = \frac{-1 + \sqrt{1+4A}}{2A}. \tag{33}$$

Then

$$\left(\frac{\partial X}{\partial \rho}\right)_T = \frac{dX}{dA} \left(\frac{\partial A}{\partial \rho}\right)_T, \tag{34}$$

with

$$\frac{dX}{dA} = \frac{1}{A\sqrt{1+4A}} - \frac{\sqrt{1+4A} - 1}{2A^2}, \tag{35}$$

and

$$\left(\frac{\partial A}{\partial \rho}\right)_T = m\Delta(T,\rho) + \rho m \left(\frac{\partial \Delta}{\partial \rho}\right)_T. \tag{36}$$

The second term contains the curvature feedback:

$$\left(\frac{\partial \Delta}{\partial \rho}\right)_T = \Delta_0(T)(1-\alpha)\left(\frac{\partial n}{\partial \rho}\right)_T, \tag{37}$$

and, from the smooth definition of n,

$$\left(\frac{\partial n}{\partial \rho}\right)_T = \exp\left(-\frac{R_{\text{abs}}}{R_0}\right)\frac{1}{R_0}\left(\frac{\partial R_{\text{abs}}}{\partial \rho}\right)_T, \tag{38}$$

where

$$\left(\frac{\partial R_{\text{abs}}}{\partial \rho}\right)_T = \frac{R}{\sqrt{R^2+R_\epsilon^2}}\left(\frac{\partial R}{\partial \rho}\right)_T. \tag{39}$$

In practice, $R(T,\rho)$ and its derivatives can be computed numerically from an IAPWS-95 implementation over a local stencil in (T,ρ), or by automatic differentiation if an AD-compatible implementation is available.

4.2 Isobaric expansion and the TMD line

At fixed pressure $p^\star$, the density is implicitly defined by $p(T,\rho) = p^\star$. Differentiating with respect to T gives

$$\left(\frac{\partial p}{\partial T}\right)_\rho + \left(\frac{\partial p}{\partial \rho}\right)_T\left(\frac{\partial \rho}{\partial T}\right)_p = 0, \tag{40}$$

so that

$$\left(\frac{\partial \rho}{\partial T}\right)_p = -\frac{(\partial p/\partial T)_\rho}{(\partial p/\partial \rho)_T}. \tag{41}$$

The isobaric expansion coefficient is

$$\alpha_p = -\frac{1}{\rho}\left(\frac{\partial \rho}{\partial T}\right)_p = \frac{1}{\rho}\frac{(\partial p/\partial T)_\rho}{(\partial p/\partial \rho)_T}. \tag{42}$$

A density maximum along a given isobar occurs where $\alpha_p = 0$, that is,

$$\left(\frac{\partial p}{\partial T}\right)_\rho = 0, \tag{43}$$

assuming mechanical stability $(\partial p/\partial \rho)_T > 0$. In the present model,

$$\left(\frac{\partial p}{\partial T}\right)_\rho = \left(\frac{\partial p_{\text{IAPWS}}}{\partial T}\right)_\rho + \left(\frac{\partial p_{\text{assoc}}}{\partial T}\right)_\rho. \tag{44}$$

The first term reproduces the TMD behaviour of the base IAPWS-95 formulation. The association term adds a correction. From

$$p_{\text{assoc}}(T,\rho) = \lambda R T m \rho^2\left(\frac{1}{X}-\frac{1}{2}\right)\left(\frac{\partial X}{\partial \rho}\right)_T, \tag{45}$$

we obtain

$$\left(\frac{\partial p_{\text{assoc}}}{\partial T}\right)_\rho = \lambda R m \rho^2 \Bigg[\left(\frac{1}{X} - \frac{1}{2}\right)\left(\frac{\partial X}{\partial \rho}\right)_T + T\left(\frac{\partial}{\partial T}\left\{\left(\frac{1}{X} - \frac{1}{2}\right)\left(\frac{\partial X}{\partial \rho}\right)_T\right\}\right)_\rho \Bigg]. \tag{46}$$

Since $X(T,\rho)$ depends on T through the Boltzmann factor $\exp(\varepsilon/RT)$ and through the curvature-driven factor $n(T,\rho)$, this term can become negative in regions where the network factor grows rapidly with decreasing temperature. This provides a mechanism for shifting the TMD line and modifying its shape relative to the base IAPWS prediction.

4.3 Heat-capacity anomaly

The isobaric heat capacity is

$$C_p = \left(\frac{\partial h}{\partial T}\right)_p, \tag{47}$$

where the molar enthalpy is

$$h = a_{\text{tot}} + Ts + \frac{p}{\rho}. \tag{48}$$

The entropy contribution from association is

$$s_{\text{assoc}} = -\left(\frac{\partial a_{\text{assoc}}}{\partial T}\right)_\rho, \tag{49}$$

and $h_{\text{assoc}} = a_{\text{assoc}} + Ts_{\text{assoc}}$. Taking the derivative at constant pressure,

$$\Delta C_{p,\text{assoc}} = \left(\frac{\partial h_{\text{assoc}}}{\partial T}\right)_p, \tag{50}$$

gives an association contribution to C_p dominated by first and second derivatives of $X(T,\rho)$ with respect to T. Because $X(T,\rho)$ is sensitive to both the bonding energy and the curvature-driven factor $n(T,\rho)$, the model naturally produces a broad peak in C_p in regions where $R(T,\rho)$ varies strongly with temperature, consistent with the observed enhancement of C_p on supercooling [5, 4]. A fitted quantitative evaluation requires numerical differentiation using a concrete implementation of IAPWS-95 and an optimized parameter set $(K_0, \varepsilon, \alpha, R_0, R_\epsilon, \lambda)$. Section 5 gives a preliminary finite-difference consistency check for C_p using one illustrative stable parameter set, but full reference-grade fitting remains future work.

5 Preliminary Numerical Proof-of-Concept

To check that the CDA-EOS construction is numerically executable, we carried out a preliminary grid-based demonstration using a direct-property water backend for the base EOS and the formulas above for the association contribution. This calculation is not a final fitted EOS

validation. Its purpose is narrower: verify that the base thermodynamic fields are stable, that the curvature-to-network mapping can be evaluated on a conservative single-phase grid, and that a nonzero CDA association correction can be introduced from the same free-energy term without immediately violating mechanical stability.

The base grid covered $250\,\mathrm{K} \le T \le 350\,\mathrm{K}$ and $900 \le \rho_{\mathrm{mass}} \le 1100\,\mathrm{kg\,m^{-3}}$. Non-single-phase and numerically unstable cells were masked. The direct-property backend produced the base fields p, C_p, α_p, κ_T, s, and u; on the valid base grid no cells violated $(\partial p/\partial\rho)_T > 0$, $\kappa_T > 0$, or $C_p > 0$. The density-maximum diagnostic obtained from $\alpha_p = 0$ gave $T \simeq 277.12\,\mathrm{K}$ near $\rho_{\mathrm{mass}} = 1000\,\mathrm{kg\,m^{-3}}$, consistent with the ordinary-water TMD neighbourhood.

For the curvature pass, a direct diagonal Ruppeiner metric in (T, v_m) coordinates was used:

$$g_{TT} = \frac{c_{v,m}}{RT^2}, \qquad g_{v_m v_m} = -\frac{1}{RT}\left(\frac{\partial p}{\partial v_m}\right)_T, \tag{51}$$

where $v_m = 1/\rho$ is the molar volume. Because the backend supplies derivatives with respect to mass density, the conversion

$$\left(\frac{\partial p}{\partial v_m}\right)_T = \left(\frac{\partial p}{\partial \rho_{\mathrm{mass}}}\right)_T \left(-\frac{M_w}{v_m^2}\right) \tag{52}$$

was used. This direct-metric route avoids the nested finite-difference Hessian of $s_m(u_m, v_m)$ and gave a bounded curvature field on the masked grid. In the preliminary run, 1445 robust curvature cells were retained. The resulting curvature values ranged from approximately -0.359 to 0.128 in the chosen convention, with median -0.155. The mapped network factor had $0.0068 \lesssim n \lesssim 0.902$ and median $n \simeq 0.632$.

The association sweep then evaluated

$$\Delta(T,\rho) = K_0\left(e^{(\varepsilon/R)/T} - 1\right)\left[\alpha + (1-\alpha)n(T,\rho)\right] \tag{53}$$

for a modest grid of $(K_0, \varepsilon/R, \alpha, \lambda)$ values. For each parameter set, $X(T,\rho)$, $a_{\mathrm{assoc}}(T,\rho)$, $p_{\mathrm{assoc}}(T,\rho)$, and the pressure-derived α_p and κ_T were recomputed. Of 512 tested parameter combinations, 329 retained zero mechanical-stability failures under the screening criteria. One illustrative stable parameter set is summarized in Table 1. The TMD entry is reported as a grid diagnostic rather than as a fitted TMD curve, because the present calculation is a finite-difference proof-of-concept rather than a converged reference-quality fit.

As a further response-function check, the same illustrative parameter set was used to construct a prototype heat-capacity calculation from the assembled total enthalpy. The association entropy was computed as

$$s_{\mathrm{assoc}} = -\left(\frac{\partial a_{\mathrm{assoc}}}{\partial T}\right)_\rho, \tag{54}$$

and the molar enthalpy correction was evaluated as

$$h_{\mathrm{corr}} = \lambda\left(a_{\mathrm{assoc}} + T s_{\mathrm{assoc}} + \frac{p_{\mathrm{assoc}}}{\rho}\right). \tag{55}$$

Table 1: Illustrative stable pressure-level CDA correction from the preliminary numerical sweep. This is a proof-of-concept parameter set, not a fitted final water EOS.

Quantity	Value
K_0	$1.0 \times 10^{-10}\,\mathrm{m^3\,mol^{-1}}$
ε/R	$1400\,\mathrm{K}$
α	0.5
λ	0.3
Valid pressure-derived cells	4313
Mechanical-stability failures	0
Minimum $(\partial p/\partial \rho)_T$ on valid cells	1.77×10^3 in molar-density units
Minimum κ_T on valid cells	$2.18 \times 10^{-10}\,\mathrm{Pa^{-1}}$
Median pressure correction	$0.0775\,\mathrm{MPa}$
1st to 99th percentile pressure correction	-3.94 to $0.622\,\mathrm{MPa}$
Mean TMD-diagnostic shift on common points	$-6.00\,\mathrm{K}$
TMD-diagnostic shift range on common points	-12.42 to $+1.21\,\mathrm{K}$

The prototype total heat capacity was then obtained from the grid identity

$$C_p = \left(\frac{\partial h}{\partial T}\right)_\rho - \left(\frac{\partial h}{\partial \rho}\right)_T \frac{(\partial p/\partial T)_\rho}{(\partial p/\partial \rho)_T}, \tag{56}$$

using the same total pressure surface as the mechanical-stability check. This is still a finite-difference prototype rather than a fitted EOS validation, but it tests whether the added association term can be differentiated consistently enough to produce a physically admissible C_p field. Summary results are shown in Table 2.

Table 2: Prototype C_p consistency check for the illustrative stable CDA parameter set in Table 1. Values are computed on robust masked cells using the total pressure and total enthalpy surfaces.

Quantity	Value
Robust C_p cells	4009
Cells with $C_{p,\mathrm{tot}} \leq 0$	0
Cells with $(\partial p/\partial \rho)_T \leq 0$	0
$C_{p,\mathrm{tot}}$ range	3135 to $6509\,\mathrm{J\,kg^{-1}\,K^{-1}}$
Median $C_{p,\mathrm{tot}}$	$3936\,\mathrm{J\,kg^{-1}\,K^{-1}}$
Median base C_p on same mask	$3921\,\mathrm{J\,kg^{-1}\,K^{-1}}$
Median C_p correction	$+7.1\,\mathrm{J\,kg^{-1}\,K^{-1}}$
1st to 99th percentile C_p correction	-202 to $+322\,\mathrm{J\,kg^{-1}\,K^{-1}}$
Median relative C_p change	$+0.18\%$
1st to 99th percentile relative C_p change	-5.48% to $+8.25\%$
Median base-C_p identity error	-2.7×10^{-6} relative

These results support the internal numerical feasibility of the CDA-EOS construction: the curvature-derived network factor can be evaluated on a masked liquid grid, the SAFT-

style association term can be added as a structured free-energy correction, nonzero coupling can modify the TMD diagnostic without automatically breaking mechanical stability, and a finite-difference C_p prototype can remain positive on the retained robust cells. The remaining validation task is no longer basic executability; it is reference-grade fitting and comparison of the full response-function surfaces against experimental data, IAPWS-consistent benchmarks, and existing two-state or PC-SAFT-type models.

6 Relation to Existing Water Models

Two-state and multi-state models of water have a long history, beginning with early qualitative proposals of coexisting open and collapsed local structures and extending to modern parametric models for supercooled water [4, 5]. Recent work has embedded a two-state description directly into PC-SAFT, deriving a framework (PC-SAFT-TS and PC-SAFT-TS-CAF) that reproduces the characteristic extrema of water by treating high-density and low-density water as interconverting species with fitted cross-association parameters [8, 9]. At the same time, thermodynamic curvature has been used as a diagnostic tool for fluids and for water in particular, revealing regions where the correlation length becomes large and where liquid states show solid-like character [13, 15]. These studies have largely taken an existing EOS, often IAPWS-95 or related formulations, as given and evaluated $R(T, p)$ without feeding it back into the EOS. The present proposal differs in two ways:

1. It treats IAPWS-95, or an IAPWS-consistent extension into the supercooled region [4, 17], as the fixed macroscopic base EOS and uses thermodynamic geometry of that EOS to define a network factor $n(T, \rho)$ through the curvature $R(T, \rho)$.

2. It couples this network factor into a SAFT-style association term, so that the strength of hydrogen bonding is controlled by a measure of correlation implied by the base EOS, rather than by an independent phenomenological two-state function.

In this sense the model provides a bridge between empirical high-accuracy EOS, association theory, and thermodynamic geometry. It does not attempt to replace IAPWS-95, but to augment it with an explicit network term whose intensity is tied to R.

7 Outlook

The present work is a framework with a preliminary numerical proof-of-concept, including pressure-level and prototype C_p checks, rather than a completed parametrised EOS. Several steps are needed to turn it into a practical model:

- Implement IAPWS-95 and its supercooled extension numerically and compute $R(T, \rho)$ over the region of interest, following methods similar to Refs. [13, 15].

- Choose initial guesses for $(K_0, \varepsilon, \alpha, R_0, R_\epsilon, \lambda)$ based on typical hydrogen-bond energies and known densities, then adjust them to preserve agreement with IAPWS-95 within uncertainties for $p(T, \rho)$ while modifying the predicted TMD and C_p behaviour in a controlled way.

- Refine the numerical differentiation and parameter fitting, then compare the resulting TMD line, C_p ridge, and compressibility anomalies with those from existing two-state PC-SAFT-type models [8, 9, 11] and with experimental data in the supercooled region [4, 5].

If successful, this curvature–driven association EOS would offer a way to embed network physics into an established engineering EOS, with all thermodynamic properties still derived from a single Helmholtz free-energy function.

A Numerical evaluation of curvature and derivatives

This paper treats $R(T, \rho)$ as an input computed from the IAPWS-95 Helmholtz function. The algebra in the main text assumes that $R(T, \rho)$ and at least its first derivatives with respect to (T, ρ) can be obtained in a numerically stable way. Because R involves second derivatives and nonlinear combinations of them, numerical noise can easily dominate unless the evaluation is smoothed. A practical route that stays close to the CDA-EOS needs is summarized in Fig. 4 and described below.

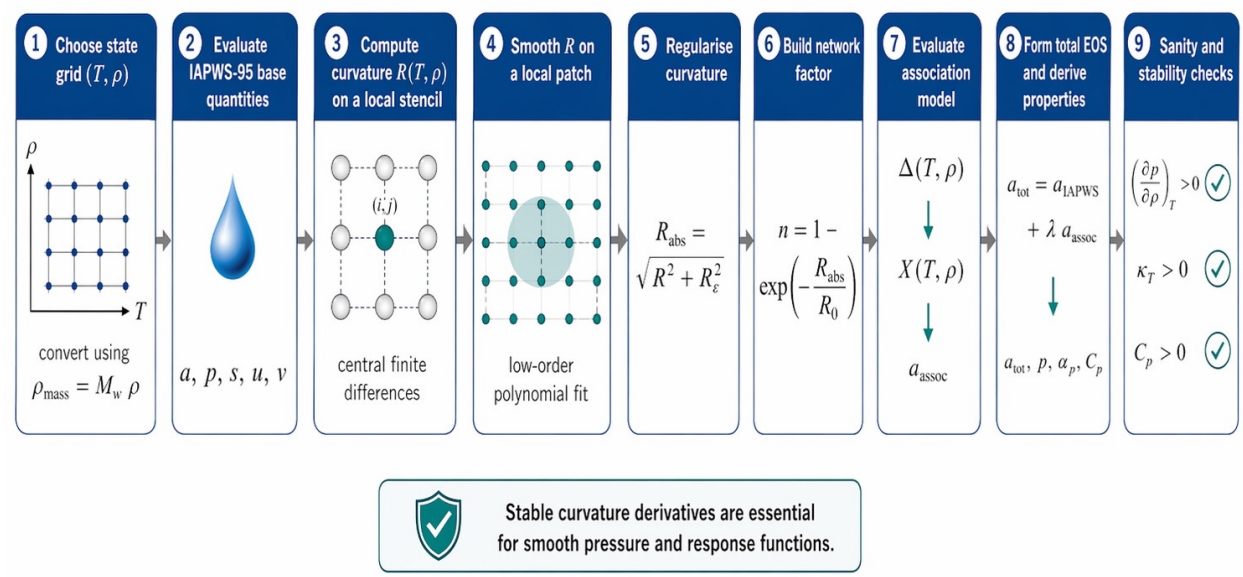

Figure 4: Numerical workflow for computing curvature, association corrections, and thermodynamic properties in the CDA-EOS. The workflow emphasizes the practical sequence needed for implementation: choose a state grid, evaluate IAPWS-95 base quantities, compute and smooth $R(T, \rho)$, regularise the curvature magnitude, build the network factor, evaluate the association model, derive thermodynamic properties, and apply sanity checks.

A.1 State grid and stable base derivatives

Work in (T, ρ) with ρ molar density and $\rho_{\text{mass}} = M_w \rho$. At each state, compute the base IAPWS quantities from $a_{\text{IAPWS}}(T, \rho)$:

$$p, \qquad s = -\left(\frac{\partial a}{\partial T}\right)_\rho, \qquad u = a + Ts, \qquad v = \frac{1}{\rho}. \tag{57}$$

Also compute base response functions that can be sanity-checked:

$$\left(\frac{\partial p}{\partial \rho}\right)_T > 0, \qquad \kappa_T = \frac{1}{\rho}\left(\frac{\partial \rho}{\partial p}\right)_T > 0, \qquad C_p > 0. \tag{58}$$

These checks catch step-size problems early.

A.2 Direct metric route used in the preliminary numerical pass

For the preliminary numerical pass in Section 5, the curvature was evaluated through a direct diagonal Ruppeiner metric in (T, v_m) coordinates rather than by repeatedly differentiating $s_m(u_m, v_m)$ on nested stencils. The metric components used were

$$g_{TT} = \frac{c_{v,m}}{RT^2}, \qquad g_{v_m v_m} = -\frac{1}{RT}\left(\frac{\partial p}{\partial v_m}\right)_T. \tag{59}$$

This route is convenient when the numerical backend directly supplies c_v and $(\partial p/\partial \rho_{\text{mass}})_T$. It still requires conservative phase masking and boundary masking, but in the preliminary calculation it produced a substantially smoother working curvature field than the more brute-force entropy-Hessian route.

A.3 Finite differences for R and its derivatives

Use a local symmetric stencil around (T, ρ):

$$T_\pm = T \pm \Delta T, \qquad \rho_\pm = \rho \pm \Delta\rho, \tag{60}$$

with relative step sizes chosen small but not tiny:

$$\Delta T = \eta_T T, \qquad \Delta\rho = \eta_\rho \rho, \tag{61}$$

where typical starting values are $\eta_T, \eta_\rho \sim 10^{-3}$ to 10^{-4}. In strongly varying supercooled regions one may need to increase steps to suppress noise. Compute R at the central state and neighbours using one consistent procedure. Then compute first derivatives with central differences:

$$\left(\frac{\partial R}{\partial T}\right)_\rho \approx \frac{R(T_+, \rho) - R(T_-, \rho)}{2\Delta T}, \tag{62}$$

$$\left(\frac{\partial R}{\partial \rho}\right)_T \approx \frac{R(T, \rho_+) - R(T, \rho_-)}{2\Delta\rho}. \tag{63}$$

If the derivative flips wildly when (η_T, η_ρ) is changed by a factor of 2, the calculation is not stable yet and needs smoothing.

A.4 Local smoothing of R

A simple way to smooth while keeping locality is to fit a low-order polynomial surface to R values on a small (T, ρ) patch, then differentiate the polynomial. For example, fit on a 3×3 or 5×5 patch:

$$R(T, \rho) \approx \sum_{i=0}^{2} \sum_{j=0}^{2} c_{ij} (T - T_0)^i (\rho - \rho_0)^j, \tag{64}$$

solve for c_{ij} by least squares, then take analytic derivatives of the fit. This tends to reduce stencil noise without washing out real structure.

A.5 Regularised magnitude used in the CDA-EOS

Once R is stable, use the regularised magnitude already defined in the main text:

$$R_{\text{abs}} = \sqrt{R^2 + R_\epsilon^2}, \qquad \left(\frac{\partial R_{\text{abs}}}{\partial \rho}\right)_T = \frac{R}{\sqrt{R^2 + R_\epsilon^2}} \left(\frac{\partial R}{\partial \rho}\right)_T. \tag{65}$$

This avoids singular behaviour of $n(T, \rho)$ derivatives at $R = 0$.

A.6 What the CDA-EOS actually needs

For the pressure correction p_{assoc}, the curvature enters through

$$\left(\frac{\partial \Delta}{\partial \rho}\right)_T \propto \left(\frac{\partial n}{\partial \rho}\right)_T \propto \left(\frac{\partial R_{\text{abs}}}{\partial \rho}\right)_T. \tag{66}$$

So the minimum viable curvature pipeline for implementation is

$$R(T, \rho), \qquad \left(\frac{\partial R}{\partial \rho}\right)_T, \tag{67}$$

computed stably enough that the resulting $p_{\text{assoc}}(T, \rho)$ is smooth.

A.7 Practical backend choices for numerical demonstrations

The preferred final implementation of the CDA-EOS is an IAPWS-95 Helmholtz backend that exposes the required reduced Helmholtz derivatives. Such a backend gives direct control over the chain-rule conversion between the native mass-density variable ρ_{mass} and the molar-density variable ρ used in this paper.

For early numerical demonstrations, however, a direct-property backend, for example CoolProp, can be used to evaluate real water fields on a (T, ρ_{mass}) grid before a full derivative-exposing IAPWS implementation is available. In this mode the backend supplies p, C_p, α_p, κ_T, s, and u directly. Those fields are sufficient for checking the state grid, masking non-single-phase cells, and verifying the stability conditions listed above.

Curvature can be estimated in such a demonstration either through the direct metric route in (T, v_m) described above or through a finite-difference route in the entropy representation. The entropy-Hessian route evaluates $s_m(u_m, v_m)$ on local stencils, forms the Ruppeiner metric

$$g_{ij} = -\frac{1}{R}\frac{\partial^2 s_m}{\partial x^i \partial x^j}, \qquad (x^1, x^2) = (u_m, v_m), \tag{68}$$

and then computes the two-dimensional scalar curvature from the metric. This route is useful as a cross-check and software scaffold, but it is slower and more sensitive near saturation and critical regions than the direct-metric, analytic-derivative, or automatic-differentiation Helmholtz implementations. For that reason, curvature evaluation in early demonstrations should be optional and restricted to conservatively masked single-phase liquid grids.

A minimal software regression check is that setting $\lambda = 0$ recovers the base EOS fields exactly. In that limit the CDA-EOS reduces to $a_{\mathrm{IAPWS}}(T, \rho)$, so any numerical implementation should return the unmodified base values for p, C_p, α_p, κ_T, s, and u. This check separates backend validity from the later problem of fitting and differentiating the curvature-driven association term.

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